

CHEM 350 Examination

Organic Chemistry: Reactions and Mechanisms

Instructions:

- Answer all questions.
 - For Questions 1–4, choose the best option.
 - For Questions 5–8, write brief but precise answers.
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1. Which factor primarily determines whether an S_N1 or S_N2 mechanism will predominate in a nucleophilic substitution reaction?
 - (A) The strength of the nucleophile only
 - (B) The structure of the alkyl halide (primary, secondary, tertiary)
 - (C) The temperature of the reaction
 - (D) The polarity of the solvent exclusively

 2. In electrophilic aromatic substitution, which substituent on a benzene ring is both ortho/para-directing and deactivating?
 - (A) Methyl group ($-\text{CH}_3$)
 - (B) Nitro group ($-\text{NO}_2$)
 - (C) Halogen atoms (e.g., $-\text{Cl}$, $-\text{Br}$)
 - (D) Amino group ($-\text{NH}_2$)

 3. What is the primary product when 1-butene undergoes hydrohalogenation with HBr in the presence of peroxides (anti-Markovnikov conditions)?
 - (A) 1-bromobutane
 - (B) 2-bromobutane
 - (C) 1,2-dibromobutane
 - (D) 2-bromo-2-methylpropane

 4. Which spectroscopic technique is most useful for distinguishing between constitutional isomers based on their connectivity?
 - (A) UV-Vis spectroscopy
 - (B) Infrared (IR) spectroscopy
 - (C) Nuclear Magnetic Resonance (NMR) spectroscopy

(D) Mass spectrometry

5. Explain the concept of carbocation stability and how it influences the regioselectivity of addition reactions to alkenes. Use Markovnikov's rule as an example.
6. Describe the mechanism of the aldol condensation reaction. What conditions favor the formation of the α, β -unsaturated carbonyl product rather than just the aldol adduct?
7. Explain the difference between kinetic and thermodynamic control in organic reactions. Provide an example where reaction conditions can be tuned to favor one product over another.
8. Discuss the role of resonance stabilization in determining the acidity of organic compounds. Compare the acidity of phenol, ethanol, and acetic acid, explaining the structural features that account for their differences.